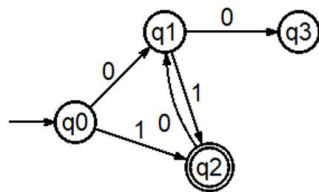


Type: MCQ

Q1. Which one of the following grammar generates the Language $L = \{a^i b^j \mid i \neq j\}$? . (0.5)

1. $S \rightarrow AC \mid CB$
 $C \rightarrow aCb \mid a \mid b$
 $A \rightarrow aA \mid \lambda$
 $B \rightarrow Bb \mid \lambda$
2. $S \rightarrow aS \mid Sb \mid a \mid b$
3. $S \rightarrow AC \mid CB$
 $C \rightarrow aCB \mid \lambda$
 $A \rightarrow aA \mid \lambda$
 $B \rightarrow Bb \mid \lambda$
4. ** $S \rightarrow AC \mid CB$
 $C \rightarrow aCB \mid \lambda$
 $A \rightarrow aA \mid a$
 $B \rightarrow Bb \mid b$

Q2. Language L given by the graph is . (0.5)



1. $L = \{x \in \{0, 1\}^* \mid x \text{ ends in 1 and does not contain substring 01}\}$
2. ** $L = \{x \in \{0, 1\}^* \mid x \text{ ends in 1 and does not contain substring 00}\}$
3. $L = \{x \in \{0, 1\}^* \mid x \text{ ends in 1 and contain substring 11}\}$
4. $L = \{x \in \{0, 1\}^* \mid x \text{ ends in 1 and does not contain substring 10}\}$

Q3. Determine the sum of minimum and maximum number of final states for a DFA having 'P' states. (0.5)

1. P
2. ** P+1
3. P-1
4. P+2

Q4. Predict the minimum number of states required in an Automaton to accept the following language. $L = \{w \mid w \text{ ends with 00}\}$.(0.5)

1. 3
2. 2
3. 4
4. ** Cannot be said

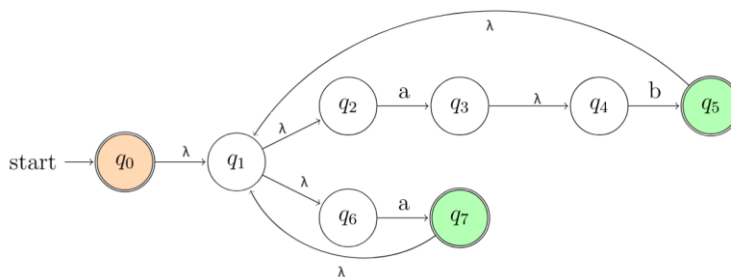
Q5. In the λ -NFA, $M = (\{q_0, q_1, q_2, q_3\}, \{a\}, \delta, q_0, \{q_3\})$ where 'S' is given in the transition table below, what is the minimum length of string to reach to the final state?

	λ	a
q_0	$\{q_1\}$	$\emptyset$
q_1	$\{q_2\}$	$\emptyset$
q_2	$\emptyset$	$\{q_2, q_3\}$
q_3	$\emptyset$	$\emptyset$

(0.5)

1. 0
2. 2
3. ****** 1
4. 3

Q6. Given the λ -NFA. What are the states reachable when 'a' is read? (0.5)



1. $\{q_0, q_1, q_2, q_3, q_4, q_5, q_6, q_7\}$
2. ****** $\{q_1, q_2, q_3, q_4, q_6, q_7\}$
3. $\{q_1, q_2, q_3, q_7\}$
4. $\{q_3\}$

Q7. Which one of the following are NOT equivalent regular expressions? (0.5)

1. $(01+10)^* 01$ and $((01)^*+(10)^*)^* 01$
2. $(01+10)(11)^*$ and $(01(11)^*+10(11)^*)$
3. $((01)^*+(10)^*)^*$ and $((01)^*(10)^*)^*$
4. ****** $((01)^*+(10)^*)(11)^*$ and $(01(11)^*+10(11)^*)^*$

Q8. Identify the regular expression (RE) that accepts all base 3 numbers 'b' such that $b \% 3 = 1$.

(0.5)

1. $(0 + 1 + 2)^* 0$
2. ****** $(0 + 1 + 2)^* 1$
3. $(0 + 1 + 2)^* 2$
4. $(0 + 1 + 2)^* 22$

Q9. Identify the incorrect statement. For all regular languages: (0.5)

1. A corresponding regular expression can be obtained.
2. A left linear grammar can be written.
3. A nondeterministic automaton can be drawn.

4. ** None of the mentioned

Q10. Which among the following cannot be accepted by a regular grammar? (0.5)

1. L is a set of numbers divisible by 2
2. ** L is a set of $0^n 1^n$
3. L is a set of string with odd number of 0
4. L is a set of binary complement

Type: DES

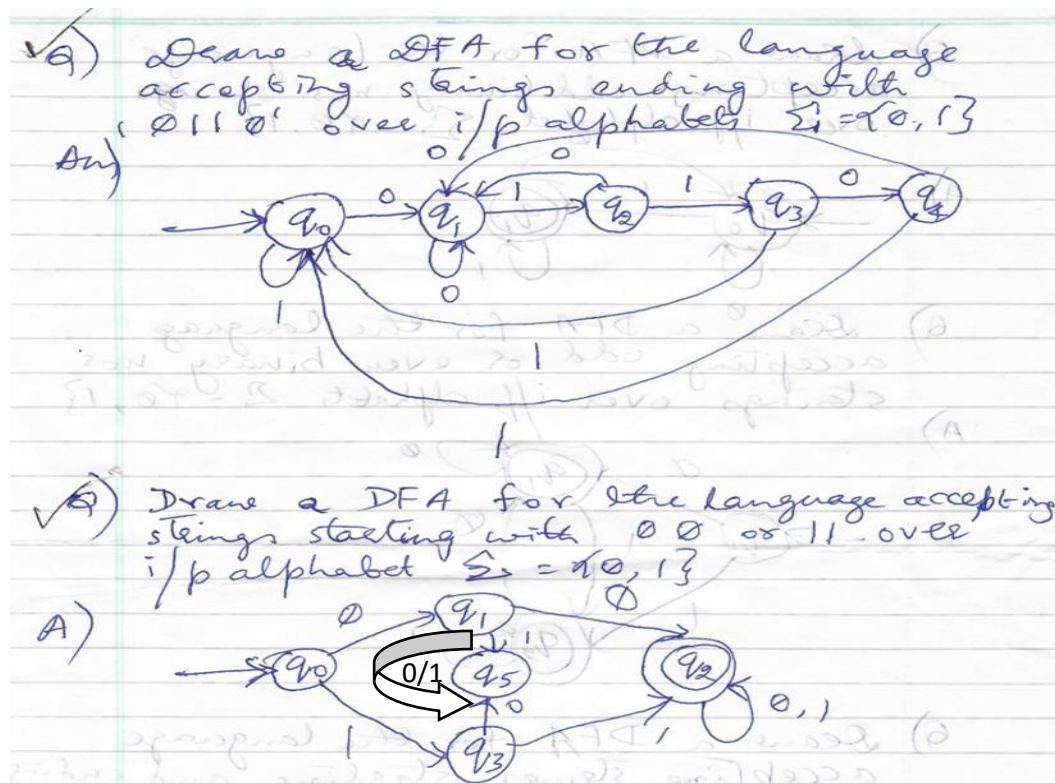
Q11. Apply the principles of DFA and draw the DFA with 5 states for the language accepting strings

i) ending with "0110" over the input alphabet $\Sigma = \{0, 1\}$

ii) starting with "00" or "11" over the input alphabet $\Sigma = \{0, 1\}$. (3)

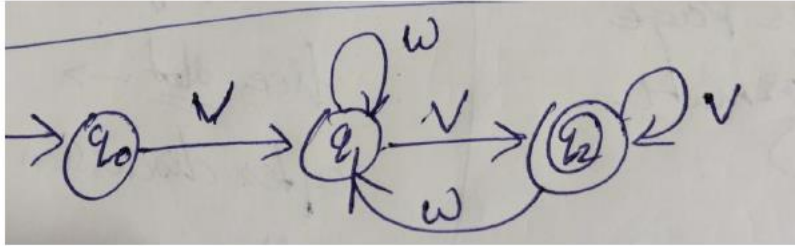
Solun: i) 1.5 M ii) 1.5 M Total -3 Marks

0/1



Q12. Show that the $L = \{vww : v, w \in \{a, b\}^*, |v| = 2\}$ is regular. (3)

Solu:



This the DFA .[Students can consider v as a or b, or a itself ,If corresponding DFA is drawn or Regular expression is given or Regular Grammar is given ,then full marks can be awarded]

$Q_0 \rightarrow VQ_1$

$Q_1 \rightarrow vQ_2$

$Q_1 \rightarrow wQ_1$

$Q_2 \rightarrow wQ_1$

$Q_2 \rightarrow vQ_2$

$Q_0 \rightarrow$ Initial state , $Q_2 \rightarrow$ Final State .

[Each steps 0.5 marks for grammar can be considered Or else if grammar is correct with less steps or correct Regular expression is given then also marks can be awarded.]

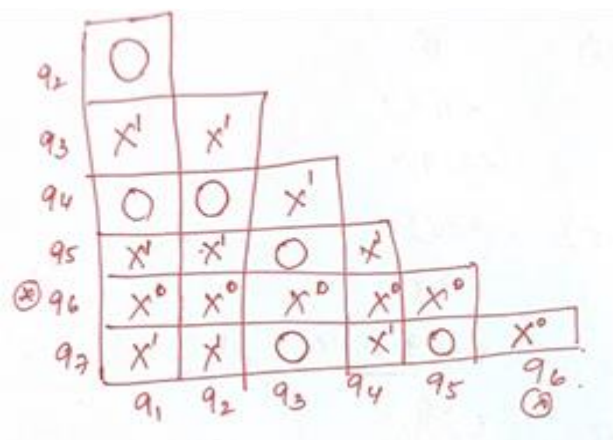
Q13. Reduce the Automaton given that q_1 is the start state and q_6 is the final state.

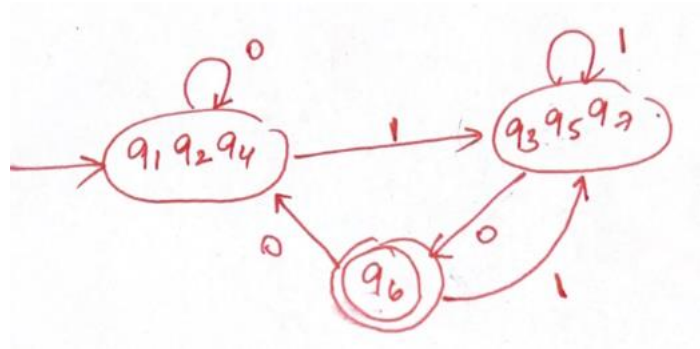
δ	0	1
q_1	q_2	q_3
q_2	q_4	q_5
q_3	q_6	q_7
q_4	q_4	q_5
q_5	q_6	q_7
q_6	q_4	q_5
q_7	q_6	q_7

(3)

Solun:.

δ	0	1
q_1	q_2	q_3
q_2	q_4	q_5
q_3	q_6	q_7
q_4	q_4	q_5
q_5	q_6	q_7
q_6	q_4	q_5
q_7	q_6	q_7



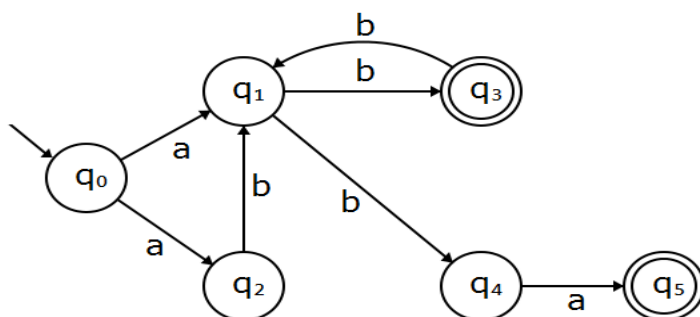


1 Mark for the table and marking final states.

1 Mark for distinguishable states

1 Mark for Minimized automata after merging states

Q14. For the NFA given in figure, Give the transition table for the equivalent DFA using Subset Construction Method



. (3)

Solun:

	a	b
$\rightarrow \{q_0\}$	$\{q_1, q_2\}$	$\emptyset$
$\{q_1, q_2\}$	$\emptyset$	$\{q_1, q_3, q_4\}$
$\{q_1, q_3, q_4\}$	$\{q_5\}$	$\{q_3, q_4, q_1\}$
$\{q_5\}$	$\emptyset$	$\emptyset$
$\emptyset$	$\emptyset$	$\emptyset$

1 mark for the table generation. 2 mark for the transitions.

Q15. Find a left linear grammar for the language generated by the grammar given below where 'S' is the start symbol. Write all the steps.

$S \rightarrow 0S \mid 1A$

$A \rightarrow 0B \mid 1C \mid \lambda$

$B \rightarrow 0S \mid 1A \mid \lambda$

$C \rightarrow 0B \mid 1C \mid \lambda$. (4)

Solun:

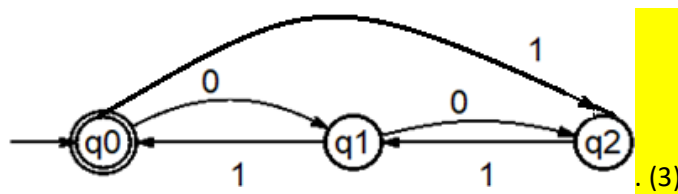
Step 1: Draw corresponding DFA

Step 2: Reverse DFA

Step 3: Obtain RLG from reversed DFA

Step 4: Obtain LLG from RLG

Q16. Derive a regular expression for the given generalised transition graph using reduction of states method



Solun:

4 steps 0.5 each

Q17. Prove that the following language is not regular. $L = \{ a^{n!} : n \geq 1 \}$ $\Sigma = \{a\}$.(3)

Solun: (String =1M ,Pumping Lemma =1M,Contracdiction =1M)

Use the Pumping Lemma. Assume for contradiction that L is a regular language. Since L is infinite. We can apply the Pumping Lemma. Let m be the integer in the Pumping Lemma. Pick a string w such that: $w \in L$, $\text{length } |w| \geq m$, we pick $w = a^{m!}$

Write $a^{m!} = x y z$

From the Pumping Lemma it must be that $\text{length } |x y| \leq m$, $|y| \geq 1$

$$xyz = a^{m!} = \underbrace{a \dots a}_x \underbrace{a \dots a}_y \underbrace{a \dots a}_{m!-m} \underbrace{a \dots a}_z$$

Thus: $y = a^k$, $1 \leq k \leq m$

$$x y z = a^{m!} \quad y = a^k, \quad 1 \leq k \leq m$$

From the Pumping Lemma: $x y^2 z \in L$

$$xy^2z = \underbrace{a \dots a}_x \underbrace{a \dots a}_{m+k} \underbrace{a \dots a}_y \underbrace{a \dots a}_{m!-m} \underbrace{a \dots a}_z \in L$$

Thus: $a^{m!+k} \in L$

$$x y z = a^{m!} \quad y = a^k, \quad 1 \leq k \leq m$$

From the Pumping Lemma: $x y^i z \in L$

$$i = 0, 1, 2, \dots$$

Thus: $x y^2 z \in L$

$$a^{m!+k} \in L \quad 1 \leq k \leq m$$

Since: $L = \{a^{n!} : n \geq 0\}$



There must exist p such that:

$$m!+k = p!$$

However: $m!+k \leq m!+m$ for $m > 1$ $a^{m!+k} \in L$ $1 \leq k \leq m$

$$\begin{aligned} &\leq m!+m! \\ &< m!m + m! \\ &= m!(m+1) \\ &= (m+1)! \end{aligned}$$

↓

$$m!+k < (m+1)!$$

↓

$$m!+k \neq p! \quad \text{for any } p$$

BUT: $L = \{a^{n!} : n \geq 0\}$

↓

$a^{m!+k} \notin L$

CONTRADICTION!!!

Therefore: Our assumption that L is a regular language is not true.
Conclusion: L is not a regular language

Q18.(i) Show that the grammar is ambiguous. Consider string $w = 0101011$

$$S \rightarrow 0/01S1/0A1$$

$$A \rightarrow 1S/0AA1 \quad . (2)$$

Solu: Two Leftmost Derivations:

Two Leftmost Derivations:

$$\begin{aligned} (1) \quad S &\rightarrow 01S1 \\ &\rightarrow 010A11 \\ &\rightarrow 0101S11 \\ &\rightarrow 0101011 \end{aligned}$$

$$\begin{aligned} (2) \quad S &\rightarrow 01S1 \\ &\rightarrow 0101S11 \\ &\rightarrow 0101011 \end{aligned}$$

There are two different leftmost derivative hence the language is ambiguous.

Q18.(ii) Find context free grammar for the following language with $n \geq 0$ and $m \geq 0$

i. $L = \{a^n b^m : 2n \leq m \leq 3n\} \quad . (2)$

Solu:

n	m	$n \leq m \leq 2n$ condition	string
0	0	$a^0 b^0$	ϵ
1	1	$a^1 b^1$	ab
1	2	$a^1 b^2$	abb
2	2	$a^2 b^2$	$aabb$
2	3	$a^2 b^3$	$aabbb$

$$\begin{aligned} S &\rightarrow \epsilon \\ S &\rightarrow aSb \\ S &\rightarrow aSbb \end{aligned}$$

$$\begin{aligned} G &= (V, \Sigma, R, S) \\ &= (\{S, a, b\}, \{a, b\}, R, S) \text{ where } R = \{S \rightarrow aSb \mid aSbb \mid \epsilon\} \end{aligned}$$